
DATA VISUALIZATION PROJECT MEMO

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Subject: UNM Enrollment Data – Visually Conceptualized

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Objective

The purpose of this memo is to represent data from the UNM 2019 Enrollment findings using four different chart visualizations. Each chart fits a different need, which we explain more through each section. Lastly, our tool for this project was Tableau.

Summary

The University of New Mexico Enrollment Data serves as enrollment verification for the census, funding, and research. These reports are used by other organizations like the Department of Education for financial aid services and funding allocation. We are representing a small portion of that information here.

The data we are representing is the average student workload. We are using the average student workload of each school or college and comparing it to data from other years. This will help students who are interested in continuing education, understand the workload of each degree concentration. Because the data is transparent, it will better help them compare their options. medical school over a different college.

Solution Average Student Workload

We chose to present this data in four different charts: table, bar graph, line graph, and horizontal bar graph in order to help students see the variances in UNM concentrations. We chose a table because it's the base for our data because it logically stores all the values being represented. We chose the line graph because it highlights the workload trend across each college or school. We chose the bar graph because it does that of a line graph but emphasizes discretely the work load of each individual college or school. Lastly, the horizontal bar graphs emphasize from largest to smallest the workload of each college or school.

TABLE 1: AVERAGE STUDENT WORKLOAD PER COLLEGE PER YEAR

Table Graph:

This table showcases the average student workload from a given school or college between the year of 2015 to 2019. This graph signals great information because it provides students an overview of the average workload expected of them for the school or college. This table, when graphed, represents the average student workload over a year to 2019. Therefore, we excluded that data from the table because the values we are representing those values in a different mode. For instance, the the change from 1 to 5 years were percentages and everything else was an integers so we could not graph the percentages. Therefore, by reformatting the data set, we made the table easier to read.

Average Student Work Load

Colleges/ Schools	2015	2016	2017	2018	2019
Anderson Graduate	7.84	8.00	7.69	7.78	7.52
Concurrent	4.13	4.00	4.21	3.72	2.99
Doctor of Nursing Practice (DNP)	6.60	7.00	6.35	5.24	5.67
Doctor of Pharmacy (PHARM.D)	17.48	19.00	18.18	16.87	17.56
Doctor of Physical Therapy (DPT)	17.00	17.00	17.32	17.35	17.23
Graduate Programs	7.88	8.00	7.83	7.79	7.71
Grand Total	12.27	12.00	12.21	12.12	12.07
Law School	14.12	14.00	13.96	14.09	13.53
Medical School	22.18	25.00	25.34	26.09	25.00
Non-Degree Graduate	4.55	5.00	4.31	4.43	4.37
Non-Degree Undergraduate	10.73	11.00	10.51	10.42	10.40
Undergraduate	13.47	14.00	13.42	13.33	13.27

2015, 2016, 2017, 2018 and 2019 broken down by Colleges/ Schools.

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TABLE 2: AVERAGE STUDENT WORKLOAD PER COLLEGE PER YEAR

Line Graph:

This line graph was conceptualized to showcase the trend in average student load by college school from 2015-2019. The graph emphasizes effectively the change in magnitude of one to the others. Thus making it more effective for students to understand the different workloads each school or college requires. Again we excluded the rate of change from 1 to 5 years because a line graph illustrates it better. Therefore, making it much easier to process the information. This, in combination with our color palette which distinguishes the different colleges and schools, creates an easy to read graph.

Average Student Work Load



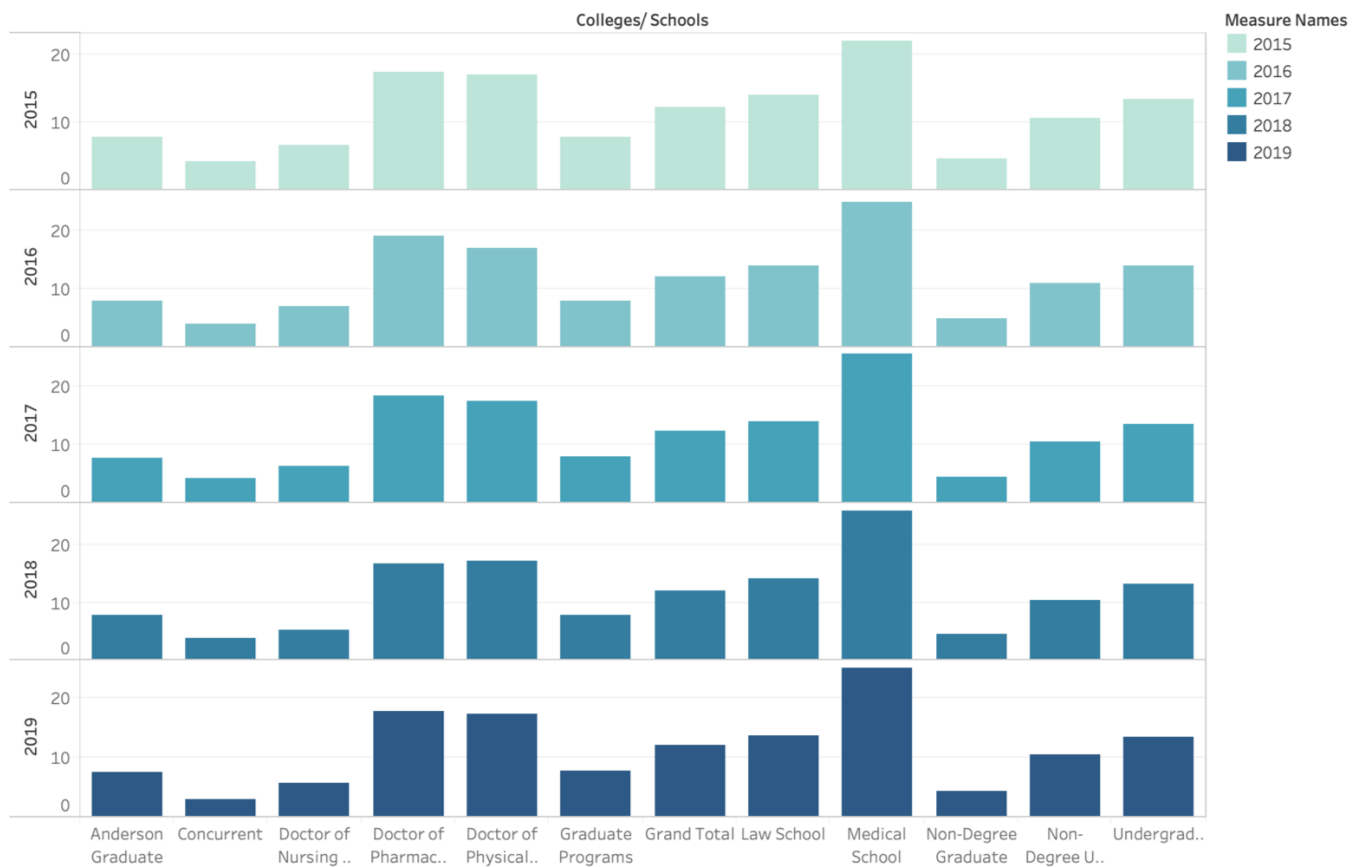
The trends of sum of 2015, sum of 2016, sum of 2017, sum of 2018 and sum of 2019 for Colleges/ Schools. Color shows details about Colleges/ Schools.

TABLE 3: AVERAGE STUDENT WORKLOAD PER COLLEGE PER YEAR

Bar Graph:

This bar graph (table 3) represents the colleges/schools as ordinal variables. Meaning that are variables follow a natural progression or range. The procession it follows is the one provided by the Institution from the table. The table which is the original data showcased the itself as numerals, but we represented it by color coded bars. Thus, illustrating the workload visually instead of numerically. This facilitates an easier experience for students to choose a degree plan and our target audience gets more use out of this visualization since it's easier to access and read.

Average Student Work Load



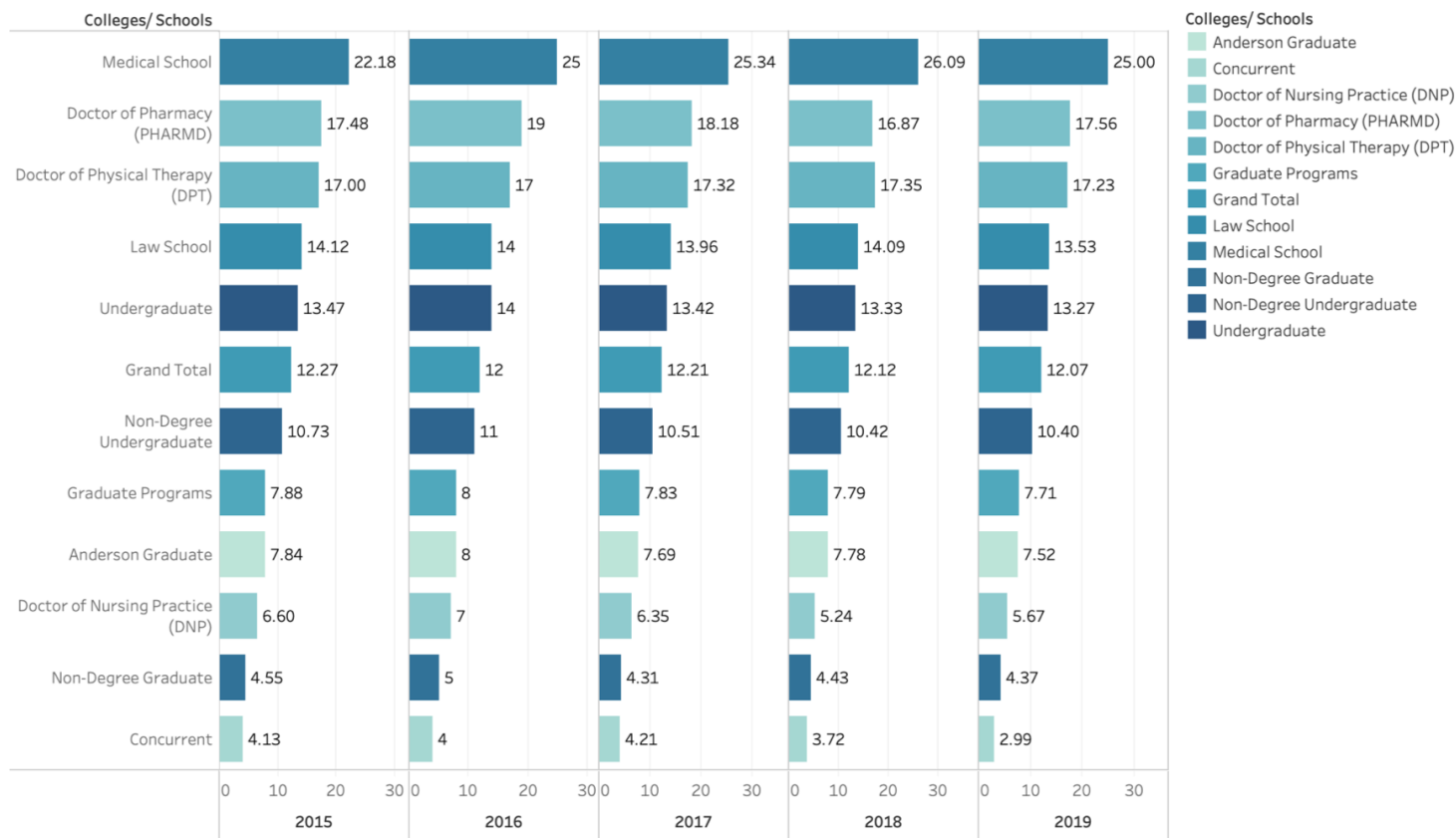
2015, 2016, 2017, 2018 and 2019 for each Colleges/ Schools. Color shows details about 2015, 2016, 2017, 2018 and 2019.

TABLE 4: AVERAGE STUDENT WORKLOAD PER COLLEGE PER YEAR

Horizontal Graph:

This horizontal graph (table 4) represents the colleges/schools as nominal variables, meaning that we are representing the different colleges/schools in an order. For example, we ordered the data from largest workload to smallest workload per college per year. Thus, emphasizing the largest average workload for a college/school instead of each college/school's workload (as shown in Table 3). This signals a better understanding of each college/school's workload and paints a better picture for their potential concentration.

Average Student Work Load



Sum of 2015, sum of 2016, sum of 2017, sum of 2018 and sum of 2019 for each Colleges/ Schools. Color shows details about Colleges/ Schools.

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Conclusion:

We generated these four visual representations (a table, a bar graph, a line graph, horizontal graph) from the Office of Institutional Analytics data. We specifically focused on the average student workload and found that Medical degrees on average have a higher student workload. Surprisingly, this information highlights that, in comparison to other disciplines, medicine is most popular at UNM.
